Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Among the following ions, the $p\pi - d\pi$ overlap could be present in

(A) NO_3^-

(B) PO_4^{3-}

(C) CO_3^{2-}

(D) NO_2^-

Q.2

The bond dissociation energy of $B - F$ in BF_3 is $646\ kJ\ mol^{-1}$ whereas that of $C - F$ in CF_4 is $515\ kJ\ mol^{-1}$. The correct reason for higher $B - F$ bond dissociation energy as compared to that of $C - F$ is

(A) stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4.

(B) significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4.

(C) lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4.

(D) smaller size of B-atom as compared to that of C -atom.

Q.3

In PO_4^{3-} ion the formal charge on the oxygen atom of $P - O$ bond is ___

(A) 0.75

(B) -0.65

(C) -0.75

(D) -0.69

Q.4

Which of the following contains both covalent and ionic bond?

(A) NH_4Cl

(B) H_2O

(C) CCl_4

@IITJEE_Advanced

(D) $CaCl_2$

Q.5

Which of these statements is not true?

(A) NO^+ is isoelectronic with O_2

(B) B is always covalent in its compounds

(C) In aqueous solution, the Tl^+ ion is much more stable than Tl (III)

(D) $LiAlH_4$ is a versatile reducing agent in organic synthesis.

Q.6

Which of the following does not apply to metallic bond ?

(A) Overlapping valence orbitals

(B) Mobile valency electrons

(C) Delocalized electrons

(D) Highly directed bonds.

Q.7

Which of the correct increasing order of lone pair of electrons on the central atom?

(A) $IF_7 < IF_5 < ClF_3 < XeF_2$

(B) $IF_7 < XeF_2 < ClF_2 < IF_5$

(C) $IF_7 < ClF_3 < XeF_2 < IF_5$

(D) $IF_7 < XeF_2 < IF_5 < ClF_3$

Q.8

The molecules BF_3 and NF_3 are both covalent compounds, but BF_3 is non polar whereas NF_3 is polar. The reason for this is

(A) atomic size of boron is larger than nitrogen

(B) Boron is metal while nitrogen is gas

(C) $B - F$ bonds are non-polar while $N - F$ bonds are polar

(D) BF_3 is planar but NF_3 is pyramidal

Q.9

The number and type of bonds in ion in CaC_2 are:

@IITJEE_Advanced

(A) One σ bond and one π -bond

(B) One σ bond and two π -bond

(C) Two σ bond and two π -bond

(D) Two σ bond and one π -bond

Q.10

In XeF_2, XeF_4 and XeF_6, the number of lone pairs on Xe are respectively

(A) $2, 3, 1$

(B) $1, 2, 3$

(C) $4, 1, 2$

(D) $3, 2, 1$

Q.11

Which of the following molecules has two sigma (\sigma) and two pi (π) bonds?

(A) C_2H_4

(B) $N_2\ F_2$

(C) $C_2H_2Cl_2$

(D) HCN

Q.12

The maximum number of H-bonds in which water molecule can participate is

(A) 1

(B) 3

(C) 2

(D) 4

Q.13

Which has a coordinate bond

(A) SO_3^{2-}

(B) CH_4

(C) CO_2

(D) NH_3

@IITJEE_Advanced

Q.14

Which of the following has least covalent $P-H$ bond?

(A) PH_6^+

(B) P_2H_6

(C) P_2H_5

(D) PH_3

Q.15

Which of the following pairs of species are not isostructural?

(A) XeF_4, PF_4^+

(B) CH_4 and NH_4^+

(C) XeF_2, I_3^-

(D) XeF_5^+, IF_5

Answers & Solutions

Q.1 Answer:

PO_4^{3-}

Solution:

In $P=O$ bond, π - bond is formed by the sidewise overlapping of d - orbital of P and p - orbital of oxygen. Hence, it is formed by $p\pi$ and $d\pi$ overlapping.

In nitrogen and carbon, no vacant d-orbital is present. So, they do not form $p\pi - d\pi$ bond.

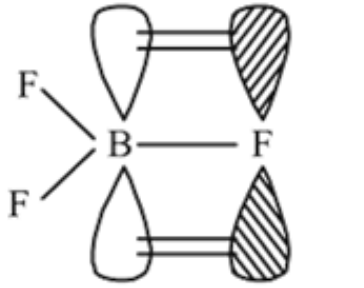
Q.2 Answer:

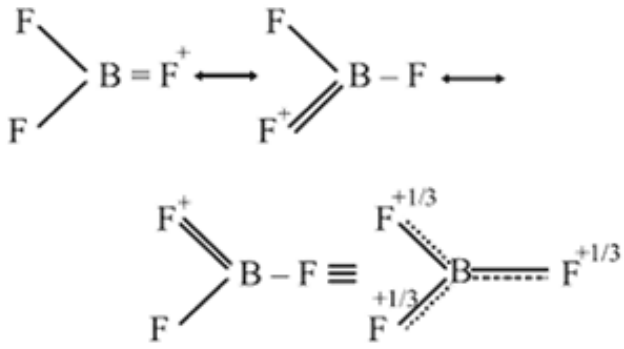
significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4.

Solution:

The delocalised $p\pi - p\pi$ bonding between filled p -orbital of F and vacant p -orbital of B leads to shortening of $B-F$ bond length which results in higher bond dissociation energy of the B-F bond.

@IITJEE_Advanced





Q.3 Answer:

-0.75

Solution:

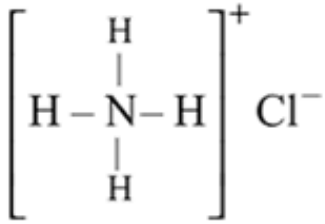
In PO_4^{3-} ion, formal charge on each O -atom of $P-O$ bond $= \frac{\text{Total charge}}{\text{Number of O atom}} = -\frac{3}{4} = -0.75$

Q.4 Answer:

NH_4Cl

Solution:

NH_4Cl contains both covalent and ionic

bonds 

Q.5 Answer:

NO^+ is isoelectronic with O_2

Solution:

$NO^+ = 7 + 8 - 1 = 14e^-$.

$O_2 = 16e^-$ **i.e not iso-electronic**

(b) Boron forms only covalent compounds. This is due to its extremely high ionisation energy.

@IITJEE_Advanced

(c) Compounds of Tl^{+} are much more stable than those of Tl^{3+}.

(d) $LiAlH_4$ is a versatile reducing agent in organic synthesis

Q.6 Answer:

Highly directed bonds.

Solution:

In metallic bonds, each ion is surrounded by equal number of oppositely charged electrons, hence have equal electrostatic attraction from all sides and hence do not have directional characteristics.

Q.7 Answer:

$IF_7 < IF_5 < ClF_3 < XeF_2$

Solution:

The number of lone pairs of electrons on central atom in various given species are

\begin{array}{ll}

**Species & Number of lone pairs on central atom **

**IF _{7} & nil **

**\mathrm{IF}_{5} & 1 **

**\mathrm{ClF}_{3} & 2 **

**\mathrm{XeF}_{2} & 3 **

\hline

\end{array}

Thus the correct increasing order is

$IF_7 < IF_5 < ClF_3 < XeF_2$

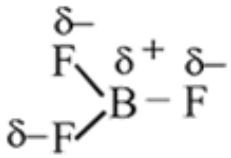
0 1 2 3

Q.8 Answer:

BF_3 is planar but NF_3 is pyramidal

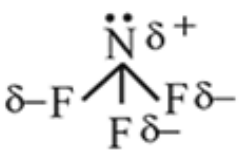
Solution:

The shape of BF_3 is trigonal planar



and $\mu = 0$ hence it is non polar.

The shape of NF_3 is pyramidal



@IITJEE_Advanced

and $\mu \neq 0$ hence it is polar.

Q.9 Answer:

One σ bond and two π -bond

Solution:

The structure of CaC_2 is $Ca^{2+}[: C \equiv C :]^{2}-$
i.e, one σ and two π bonds

Q.10 Answer:

$3, 2, 1$

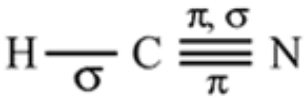
Solution:

In XeF_2 Total number of valence electrons of $Xe = 8$, two electrons shared with 2 F atoms, 6 electrons left hence 3 lone pairs, in $XeF_4 4$ shared with 4 F atoms 4 left hence 2 lone pairs; in $XeF_6 6$ shared with 6 F atoms 2 left hence 1 lone pair.

Q.11 Answer:

HCN

Solution:



i.e 2π and 2σ bonds.

Q.12 Answer:

4

Solution:

In H_2O molecule, two H-bonds are formed with O atom and two are formed with H atoms of the H_2O molecule.

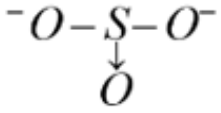
Q.13 Answer:

SO_3^{2-}

Solution:

@IITJEE_Advanced

SO_3^{2-} **has one coordinate bond.**



Q.14 Answer:

PH_6^+

Solution:

In PH_6^+, $P-H$ bond will have ionic character due to electron attracting tendency of P-carrying positive charge and least covalent.

Q.15 Answer:

XeF_4, PF_4^+

Solution:

XeF_4 **has 6 pairs of electrons around Xe while PF_4^+ has 4 pairs of electrons around P.**

@IITJEE_Advanced